

SAMRAT MALISETTI

Software Engineer — Agentic AI & LLM Systems

California, United States · samratmaliseti@gmail.com · linkedin.com/in/samrat-maliseti · github.com/samratdotm · samratmaliseti.dev

SUMMARY

Software engineer with **5+ years of experience**, currently building production AI agent systems at ADP that serve **41M+ employees across 1.1M clients**. Specialty: making LLM agents safe to trust with real actions — guardrails, human-in-the-loop flows, eval frameworks, durable memory, and observability at every layer. The more important the answer, the less the AI should improvise: models interpret the human, deterministic code does the math, evals gate every release.

EXPERIENCE

Software Engineer — ADP

May 2025 – Present

Pasadena, CA · Production AI agents on ADP Assist (41M+ employees, 1.1M clients)

- Core engineer on the **WFM Clocking Assistant** — a conversational agent that lets employees clock in/out, manage breaks, handle attestations, and complete department transfers through natural language, with compliance checks built in.
- Architected the agent on the **Strands Agent SDK** (agent loop, tool system, steering hooks), routed through ADP's AI Gateway via Azure OpenAI, with an intent orchestrator deciding where each query goes.
- Designed **human-in-the-loop confirmation flows** with hash validation, expiry, and audit logging, plus steering guardrails enforcing business rules — an agent that punches a timecard wrong is a payroll problem, not a chatbot quirk.
- Implemented durable cross-conversation memory with **AgentCore Memory on AWS Bedrock**, persisting only what's needed.
- Cut time-to-first-token cost by **75%** with a 2-layer prefix-caching strategy (global static + per-session context).
- Built an automated **eval framework** that tests agent behavior before anything ships — intent mapping, action parameters, HITL confirmations, steering blocks, and multi-turn conversations.
- Added production observability with **Langfuse**: per-request tracing of tool calls, latency, token usage, and cost.
- Reduced CI/CD deployment time from **80 minutes to 12** on Concourse + Kubernetes.

Full Stack Engineer (Contract) — GorgeousTV

Jun 2024 – Feb 2025

Los Angeles, CA · Remote

- Analyzed cloud resource usage across GCP and AWS to identify cost savings and performance improvements as part of an Azure migration initiative; evaluated multi-cloud compatibility with **Azure Migrate**.
- Configured Azure Virtual Networks, subnets, and network security groups, mapping existing GCP infrastructure to equivalent Azure architecture for seamless migration.
- Deployed containerized services with **Kubernetes + Docker**, implementing scalable deployment strategies to improve reliability and uptime.
- Automated infrastructure monitoring and logging with **Prometheus + Grafana**, improving observability and incident response.
- Managed IAM roles and access policies across GCP, AWS, and Azure, enforcing consistent access control throughout the migration.

Software Engineer — ZeMoSo Technologies

May 2021 – Aug 2022

Hyderabad, India · Field Squared field-service platform (React Native)

- Designed and implemented an **offline request queue** so field technicians could keep working without network access, syncing data automatically once connectivity was restored.
- Enabled enterprises to automate field-service operations with features for task scheduling, data collection, and workflow management; improved UI/UX from direct technician feedback.
- Optimized GraphQL middleware, cutting average latency and response times by **25%**.

Hyderabad, India

- Built a college management platform end-to-end with React, NestJS, Express, and MongoDB — RESTful APIs and database schemas designed from scratch.
- Implemented **role-based access control** per API using custom middlewares, decorators, and PassportJS authentication for multiple user roles.
- Used NestJS Guards and **WebSockets** for real-time security protocols; improved performance with lazy loading, task scheduling, and cookie management.

SELECTED PROJECTS

Amparo AI — Multilingual Insurance Voice Agent

YC Hackathon · 2026

Phone-call agent that answers health-plan questions in any language: **24 parallel retrievals at <100ms** per turn → cited 4-plan cost comparison in under 2 seconds. The LLM never computes a number — deterministic code does the math, every fact is cited to its exact location in the real plan PDF, and a regex-based clinical-handoff guardrail routes medical questions to a doctor. Exposed a "cheapest" plan actually costing **\$22k more per year** due to out-of-network rules; 85 tests. LiveKit · Python · SIP · TTS/STT.

Engram — Shared Account Brain for Sales Teams

Agentic AI Hackathon SF · 2026

iMessage agent that detects contradictions against a team fact ledger, **supersedes stale facts with full version lineage** (v1→v6), and delivers source-attributed account briefs on demand. Express/TypeScript spine with extract → reconcile → answer pipelines and a live memory panel.

Specforge — Speak an API Into Existence

Google I/O Hackathon · 2026

A Managed Gemini Agent writes a backend from plain English, **tests it in its own sandboxed VM until green**, and a FastAPI control plane hot-reloads it behind a public URL; the service evolves across conversational turns ("now add auth").

Platewise — AI Meal Planner (Android)

2026

Built in a weekend, live as a sideloadable APK: generates a personalized **7-day plan with zero recipe repeats** from cuisines, diet, and pantry via the Claude API, with one-tap recipe swaps and an auto-generated grocery list. React Native · Expo · EAS.

Stock Market Dashboard — Performance Deep Dive

2025

Virtualized infinite scroll stays smooth at **10,000+ rows** (90% fewer DOM nodes, constant memory); deferred-value search and react-query caching cut redundant API calls ~60% and initial load 40%.

SKILLS

AGENTIC AI	Strands Agent SDK, AWS Bedrock · AgentCore Memory, Azure OpenAI, Claude API, Gemini, multi-agent orchestration, RAG & hybrid retrieval, eval frameworks, LLM observability (Langfuse), HITL guardrails, prompt caching, MCP, voice agents (LiveKit · STT/TTS)
LANGUAGES	TypeScript, JavaScript, Python, Java, SQL
WEB & MOBILE	React, Next.js, React Native, Expo, Node.js, NestJS, Express, FastAPI, GraphQL, WebRTC, PostgreSQL, MongoDB
CLOUD & DEVOPS	AWS, GCP, Azure, Kubernetes, Docker, CI/CD (Concourse), Prometheus, Grafana

EDUCATION

M.S. Computer Science — California State University, Long Beach

2022 – 2024

- Open-source contributions during graduate study: SuperMemory, Polar, TensorZero — UI/UX, API integration, performance.